

PHANEENDRA GAVARA

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SUMMARY

M.S. Data Science graduate from ASU (GPA 3.90/4.0) with hands-on experience building ETL pipelines, star-schema data warehouses, hybrid RAG search engines, BI dashboards, and cloud-native data systems on **Azure**, **AWS**, and **GCP**. Proficient in Python, SQL, FastAPI, Spark, Tableau, and scalable data tooling. Targeting Data Scientist, Data Analyst, Data Engineer, and ML Engineer roles.

EDUCATION

Arizona State University

M.S. Data Science, Analytics & Engineering **GPA: 3.90 / 4.0**

- Concentration: Computing and Decision Analytics

Tempe, AZ

Aug 2024 – May 2026

IIIT Bhubaneswar

B.Tech in Computer Science Engineering

- Core CS: Algorithms & Data Structures, DBMS, Operating Systems, Computer Networks

Bhubaneswar, India

Dec 2020 – May 2024

CERTIFICATIONS

AWS Certified AI Practitioner — Amazon Web Services, Jun 2026

[Credly](#)

Anthropic (May 2026): AI Fluency Framework & Foundations · Introduction to Claude Cowork · Claude Code 101 · Claude 101

TECHNICAL SKILLS

Languages & Query: Python, SQL, R, JavaScript, TypeScript

Statistics & Analytics: Hypothesis Testing, A/B Testing, Regression, Experiment Design, Causal Inference, SARIMAX, ARIMA, Feature Engineering, SHAP, Cross-Validation

ML & Deep Learning: Scikit-learn, PyTorch, CNNs, HuggingFace Transformers, DistilBERT, NLP, Pandas, NumPy, SciPy

BI & Visualization: Tableau, Power BI, Sigma, Recharts, Matplotlib, Seaborn, Plotly

Data Engineering: Star Schema Design, ETL Pipelines, Columnstore Indexing, Apache Spark, PySpark, Kafka, PostgreSQL, Azure SQL, OR-Tools

Cloud - Azure: Blob Storage, App Service, AI Search, App Insights **AWS:** S3, EC2, SageMaker, Lambda **GCP:** (basics)

MLOps & Backend: FastAPI, Pydantic, JWT/RBAC, Bicep IaC, MLflow, Docker, REST APIs, Git, JIRA

PROJECTS

InsightHub — **AI Business Analytics Platform** | *Python, FastAPI, Azure SQL, Azure AI Search, Azure OpenAI, React* | [Live](#) Jun 2026

- Python ETL pipeline processing 230,000+ rows from 7 CSVs (Azure Blob Storage) into a star schema (6 dims, 3 fact tables); Non-Clustered Columnstore Index cut FactSales aggregation from ~8 s to <1 s on 119,652 rows
- Hybrid RAG search (BM25 + ada-002 + Azure AI Search semantic re-ranking) over 20 business docs; GPT-4o JSON-mode insights from live SQL metrics; FastAPI + JWT/RBAC backend; Bicep IaC for 13 Azure resources

TravelIQ - AI Travel Itinerary Generator | *Python, Flask, OR-Tools, scikit-learn, DistilBERT, SARIMAX, GPT-4o* May 2026

- scikit-learn Random Forest outfit recommender (weather/activity features) + FLUX.1-schnell image generation (HuggingFace Inference API, prompt-hash cached); SARIMAX crowd forecasting (20+ locations) + DistilBERT sentiment scoring on live reviews + OR-Tools VRPTW routing; Google Distance Matrix and Places APIs
- Full itinerary delivered in under 90 seconds via SSE streaming; GPT-4o feasibility checks applied to every generated itinerary; Flask multi-stage planning backend

ASU LLM Evaluation Pipeline | *Python, RAGAS, OpenAI API, Pinecone, GitHub Actions, SQLite, React* May 2026

- Automated data pipeline scoring 100 RAG chatbot responses per CI run across 6 statistical quality thresholds (faithfulness, hallucination rate, context precision, latency p95, cost/query); 50-run trend history stored in SQLite + JSON
- Live Recharts dashboard on Vercel with cost analytics, per-question breakdown table (category/difficulty/PASS-FAIL filters), and metric drift trend charts; judge cost reduced 25× via GPT-4o-mini model selection

Industrial AI Copilot | *Python, AWS Lambda, AWS S3, Next.js, SWR, Vercel*

Mar 2026

- Real-time anomaly detection across 4 sensor streams (temperature, pressure, vibration, flow rate) with sub-second alerting; automated incident lifecycle via **AWS Lambda**; telemetry archived to **AWS S3**

AI Voice Turing Test | *Python, Librosa, HuggingFace, NLP* | **3rd Place - ASU Social Bias Hackathon**

Oct 2024

- MFCC + spectral feature pipeline with NLP sentiment analysis to classify human vs. robotic calls at scale; ran A/B style demographic bias audit across population subgroups

Heart Disease Detection | *Python, Scikit-learn, Pandas, SHAP, Matplotlib*

Sep 2022

- **85% accuracy** on the Cleveland Heart Disease dataset; benchmarked Logistic Regression, Random Forest, and KNN with cross-validation, SHAP importance, and full feature engineering pipeline

EXPERIENCE

Teaching Aide & Grader

Aug 2025 – May 2026

Arizona State University - Fulton Schools of Engineering

Tempe, AZ

- Supported instruction across 4 courses: CSE 540 (Blockchain), CSE 230 (Assembly Language), CSE 180 (Computer Literacy), ASU 101; graded technical work and mentored students